NIFTY-50 Investment Intelligence Platform

Technical Report

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Property	Detail
Dataset	NSE NIFTY-50 Historical Data (Kaggle)
Period	January 2000 – April 2021
Stocks Covered	49 companies | 13 sectors
Total Records	~235,000 daily OHLCV records
Tech Stack	Python, XGBoost, scikit-learn, PyPortfolioOpt, SHAP, Streamlit
Deliverable	Interactive Streamlit web application
Project Duration	29 May 2026 – 13 June 2026
Contributers	Akarsh Garg, Ayush Rajoria
Source Code Repository	https://github.com/agcode-maker/investment-intelligence-nifty50.git

AI Open Project 2026 | June 2026

1. Introduction & Motivation

Financial markets generate enormous volumes of data daily, yet the gap between raw data and actionable investment insight remains wide for most investors. The NIFTY-50 index — comprising 49 of the largest companies on the National Stock Exchange (NSE) of India — spans 13 sectors and represents the backbone of the Indian equity market.

This project delivers a complete AI-powered investment intelligence platform that transforms over two decades of historical NIFTY-50 data into practical decision support. Unlike simple price-prediction systems, the platform integrates four analytical layers: predictive modelling, portfolio construction, risk assessment, and explainability — all accessible through an interactive 8-page Streamlit web application.

1.1 Dataset Overview

Property	Detail
Source	NSE NIFTY-50 Historical Data — Kaggle
Period	January 2000 – April 2021 (21+ years)
Stocks	49 companies with sufficient history
Sectors	13 sectors (Financial Services, IT, Consumer Goods, Pharma, Energy…)
Daily Fields	Date, Open, High, Low, Close, VWAP, Volume, Turnover
Metadata	stock_metadata.csv: Company Name, Industry, Symbol, ISIN
Total Records	~235,000 daily records | Avg 4,800 days per stock
Constraints	No live data, no APIs, no news/sentiment — OHLCV only

1.2 Platform Architecture

Module	File	Purpose
Data Layer	src/data_loader.py	CSV ingestion, price matrices, daily returns
Indicators	src/indicators.py	44 derived technical features from OHLCV
Predictor	src/predictor.py	XGBoost + RF ensemble, signal generation
Portfolio	src/portfolio.py	Mean-variance optimisation, discrete allocation
Risk	src/risk.py	VaR, CVaR, Sharpe, Sortino, stress testing
Anomaly	src/anomaly.py	Volatility spikes, extreme moves, Isolation Forest
Application	app/streamlit_app.py	8-page interactive Streamlit dashboard

2. Exploratory Data Analysis

2.1 Sector Composition

The NIFTY-50 universe spans 13 sectors. Financial Services is the most represented (8 companies), followed by IT (5), Consumer Goods (5), and Pharma (4). This cross-sector diversity is a fundamental advantage for portfolio diversification.

Sector	Companies	Characteristics
Financial Services	8	Banks, NBFCs — high beta, sensitive to rates
IT	5	Export-oriented, USD-linked revenue, low domestic risk
Consumer Goods	5	Defensive, stable demand, low volatility
Pharma	4	Regulated, export + domestic mix
Energy	4	Commodity-linked, high macro sensitivity
Metals	3	Highly cyclical, highest volatility in universe
Automobile	3	Domestic demand-driven, cyclical
Others	17	Cement, Construction, Telecom, Services, Media…

2.2 Data Quality & Coverage

All 49 stocks were validated for data completeness. Coverage over the 2010–2021 window (the primary analysis period) exceeds 95% for all included stocks — gaps align with legitimate non-trading periods for newly listed companies or index replacements. No imputation was performed; missing dates reflect genuine market closures.

2.3 Return & Volatility Characteristics

Metric	Value	Interpretation
Annual Volatility Range	23.6% – 44.4%	Wide spread across sectors
Median Annual Volatility	30.7%	Indian equities are inherently high-vol
Avg Pairwise Correlation (2010–21)	0.21	Moderate — diversification is meaningful
Mean Daily Return Skewness	-0.24	Slight negative — larger down days than up
Mean Excess Kurtosis	9.4	Fat tails — normal distribution underestimates risk
Best 10-year Return	2,972% (BAJAJFINSV)	Compounding in financial services
Median 10-year Return	36%	Wide dispersion; stock selection matters

2.4 Market Cycles Identified

Four major drawdown periods are visible across the universe. These periods were used as stress test scenarios in the Risk Assessment module.

Event	Period	Approx. Market Drawdown	Recovery
Global Financial Crisis	2008 – Mar 2009	~60–68%	~24 months
Euro Debt Crisis	H2 2011	~20–25%	~8 months
China Slowdown / Oil	H2 2015	~20–25%	~10 months
COVID-19 Crash	Jan – Apr 2020	~35–40%	~6 months (fastest on record)

The progressively faster recovery times reflect increasing institutional depth and policy response capacity in Indian markets — an insight directly visible in the rolling drawdown charts on the Risk Assessment page.

3. Feature Engineering

All 44 features are derived exclusively from OHLCV data, fully compliant with competition constraints. Feature computation is implemented in src/indicators.py and src/predictor.py.

3.1 Technical Indicator Families

Category	Features (count)	Specific Indicators
Moving Averages	6	SMA (20, 50, 200), EMA (12, 26, 50)
Momentum	5	RSI (14), Stochastic %K/%D, Williams %R, ROC (12)
Trend	4	MACD (12/26/9), MACD Signal, MACD Histogram, ADX (14)
Volatility	5	BB Upper/Lower/Width/%B (20d, 2σ), ATR (14), HV (21d)
Volume	3	OBV, Volume Ratio vs 20d SMA, VWAP Deviation
Price Ratios	5	vs SMA20/50/200, High-Low Range, Open-Close Range
Lag Returns	9	1d/5d/10d/20d returns; Rolling Std (5,10,20d); Rolling Mean (5,10d)
Calendar	4	Day of Week, Month, Quarter, Month-End flag
Signal Flags	3	Golden/Death Cross, RSI Overbought/Oversold, MACD Bullish Cross

3.2 Target Variables (Horizon = 5 Trading Days)

Target Variable	Type	Definition
Direction_5d	Binary (0/1)	1 if Close_{t+5} > Close_t, else 0
Future_Return_5d	Continuous (%)	(Close_{t+5} – Close_t) / Close_t × 100

3.3 Feature Importance

A Random Forest classifier trained on 2018–2021 data identified the top predictors by Gini importance. Short-term momentum and volatility regime features consistently dominated, outranking pure price-level indicators.

Rank	Feature	Category	Interpretation
1	Return_1d	Lag Returns	1-day momentum — strongest short-term signal
2	Rolling_Std_5	Volatility	5-day return volatility captures regime state
3	HV_21	Volatility	21-day historical vol — trend persistence proxy
4	RSI	Momentum	Overbought/oversold positioning
5	Price_vs_SMA20	Moving Avg	Short-term trend alignment
6	MACD	Trend	Medium-term trend direction & momentum
7	BB_Pct	Volatility	Price position within Bollinger envelope
8	ADX	Trend	Trend strength (non-directional)

4. Stock Predictor Engine (Mandatory Task A)

4.1 Model Architecture

The Stock Predictor Engine uses an ensemble of two complementary algorithms — XGBoost and Random Forest — to maximise predictive robustness. A separate XGBoost Regressor handles return forecasting. All three are combined in the EnsemblePredictor class (src/predictor.py).

Component	Algorithm	Task	Key Parameters
Classifier 1	XGBoost (XGBClassifier)	5-day direction	n_estimators=300, max_depth=5, lr=0.05, subsample=0.8
Classifier 2	Random Forest (RFC)	5-day direction	n_estimators=300, max_depth=8, min_samples_leaf=10
Regressor	XGBoost (XGBRegressor)	5-day return %	n_estimators=300, max_depth=4, lr=0.05
Ensemble	Probability averaging	Final signal	avg(XGB_prob, RF_prob), threshold=0.50

4.2 Signal Generation Logic

Signal	Condition	Meaning
STRONG BUY	P(up) > 0.60	High-confidence bullish
BUY	0.50 ≤ P(up) ≤ 0.60	Mild bullish edge
SELL	0.40 ≤ P(up) < 0.50	Mild bearish lean
STRONG SELL	P(up) < 0.40	High-confidence bearish

4.3 Validation Methodology

All models are evaluated using 5-fold walk-forward time-series cross-validation (TimeSeriesSplit from scikit-learn). Each fold trains on earlier data and tests on strictly later data — no lookahead bias. Standard k-fold is explicitly avoided as future observations must never appear in the training set.

4.4 Performance Results (5-fold Walk-Forward CV, all 49 stocks)

Metric	Value	Note
Directional Accuracy (mean)	52–55%	Evaluated across 49 stocks; varies by sector
Directional Accuracy (best stock)	~56%	IT and Consumer Goods stocks
Directional Accuracy (baseline)	50%	Random prediction — model beats baseline
Return Forecast MAE	~1.8–2.4%	Mean absolute error over 5-day horizon
Return Forecast RMSE	~2.5–3.2%	Root mean squared error
Precision (Up direction)	~0.54	Of predicted UPs, 54% correct
Recall (Up direction)	~0.55	Model captures 55% of actual up moves

A directional accuracy of 52–55% is economically significant in quantitative finance. Institutional quantitative strategies routinely operate with 51–54% edge — the alpha comes from consistent application across many stocks and periods, not from any single prediction.

5. Portfolio Construction Module (Mandatory Task B)

5.1 Methodology

Portfolio optimisation uses Modern Portfolio Theory (Markowitz, 1952) via the PyPortfolioOpt library. Expected returns are estimated as mean historical annual returns. The covariance matrix uses the sample covariance of daily returns, annualised. Both are computed over the 2015–2021 training period (1,566 trading days, 48 stocks with ≥85% data coverage).

5.2 Investor Profiles & Strategies

Profile	Strategy	Objective	Max Weight/Stock	Holdings
Conservative	Minimum Volatility	Lowest variance portfolio — capital preservation	10%	21
Balanced	Maximum Sharpe Ratio	Highest risk-adjusted return	15%	9
Aggressive	Maximum Return	Highest expected return under 35% vol cap	30%	4
Risk Parity	Inverse Volatility	Equal risk contribution from each stock	Unconstrained	48
Equal Weight	1/N Benchmark	Naive equal allocation — benchmark only	2.1%	48

5.3 Backtest Results (2015–2021)

Profile	Strategy	Annual Return	Volatility	Sharpe	Top 3 Holdings
Conservative	Min Volatility	5.6%	14.3%	0.04	HINDUNILVR, NESTLEIND, POWERGRID (10% each)
Balanced	Max Sharpe	22.5%	18.5%	0.95	ASIANPAINT, BAJAJFINSV, HINDUNILVR (15% each)
Aggressive	Max Return	28.3%	22.7%	1.03	ASIANPAINT, BAJAJFINSV, TITAN (30% each)
Risk Parity	Inv. Vol	10.9%	17.8%	0.33	48 stocks, weights ~2–3% each
Equal Weight	Benchmark	10.7%	18.5%	0.31	48 stocks, 2.08% each

The Balanced portfolio achieves a Sharpe ratio of 0.95 vs 0.31 for equal-weight — a 3× improvement through systematic optimisation. The Aggressive portfolio delivers a 28.3% annual return with a Sharpe of 1.03, concentrating in high-growth Consumer Goods and Financial Services names.

5.4 Discrete Share Allocation

PyPortfolioOpt's DiscreteAllocation module converts fractional weights into integer share counts for any investment amount. For ₹10,00,000 (10 Lakhs), the Balanced portfolio achieves near-perfect allocation with under ₹5,000 in uninvested cash. The Portfolio Builder page includes an interactive slider allowing any investment amount from ₹1L to ₹1Cr.

6. Risk Assessment Module (Mandatory Task C)

6.1 Per-Stock Risk Metrics (src/risk.py)

Metric	Formula	Interpretation
CAGR	(P_T/P_0)^(1/T) – 1	Compound annual growth rate
Annual Volatility	σ(daily ret) × √252	Total annualised price variability
Sharpe Ratio	(CAGR – Rf) / Ann.Vol [Rf=5%]	Return per unit of total risk
Sortino Ratio	(CAGR – Rf) / Downside Vol	Return per unit of downside risk only
Calmar Ratio	CAGR / |Max Drawdown|	Return vs worst historical loss
Max Drawdown	Max peak-to-trough % loss	Worst loss from any historical peak
VaR 95% (Historical)	5th percentile of daily returns	Max daily loss 95% of the time
CVaR 95%	Mean(returns ≤ VaR 95%)	Expected loss on worst 5% of days
Beta	Cov(stock,market)/Var(market)	Market sensitivity
Risk Classification	Score from Vol + MaxDD	LOW / MEDIUM / HIGH / VERY HIGH

6.2 Sample: RELIANCE Risk Profile

Metric	Value
CAGR	10.19% p.a.
Annual Volatility	38.15%
Sharpe Ratio	0.136
Sortino Ratio	0.157
Max Drawdown	-79.01%
VaR 95% (daily)	-3.08%
CVaR 95% (daily)	-5.19%
Risk Classification	VERY HIGH

6.3 Portfolio Stress Testing

Each portfolio is back-tested against historical stress periods to evaluate resilience. Results confirm that the Balanced and Aggressive portfolios' heavy allocation to Consumer Goods and IT created resilience even during COVID — these sectors benefited from the digital acceleration narrative.

Event	Period	Conservative	Balanced	Aggressive
China Slowdown	H2 2015	+2.1%	+4.1%	+5.8%
COVID-19 Crash	Jan–Jun 2020	-12.3%	-1.7%	+3.2%
COVID Recovery	Apr–Dec 2020	+38.2%	+57.1%	+71.4%

6.4 Risk Contribution Analysis

Marginal risk contribution is computed as $w_i \times (\Sigma w)_i / \sqrt{w'\Sigma w}$, decomposing total portfolio variance by stock. In the Balanced portfolio, the top 3 holdings (ASIANPAINT, BAJAJFINSV, HINDUNILVR at 15% each) contribute approximately 55–60% of total portfolio variance despite representing 45% of capital — typical of concentrated high-Sharpe portfolios. The Risk Assessment page visualises this interactively.

7. Market Anomaly Detection (Optional Task C)

The anomaly detection module (src/anomaly.py) identifies three types of unusual market events using both statistical thresholds and a machine learning approach.

7.1 Detection Methods

Method	Technique	Trigger Condition	New Columns
Volatility Spikes	Rolling z-score on 21-day realised vol	Z-score > 2.5σ above 63-day rolling mean	Vol_Zscore, VolSpike
Extreme Drawdowns	Single-day return threshold	Daily return < –5% OR Cumulative DD < –20%	DailyReturn, ExtremeDrawdown, DeepDrawdown
Volume Anomalies	Rolling z-score on daily volume	Volume Z-score > 2.5σ above 20-day SMA	Volume_Zscore, VolumeSpike
Isolation Forest	Multivariate unsupervised ML	2% contamination rate (most isolated points)	IF_Anomaly, IF_Score

7.2 Results Across Key Stocks

Stock	Total Days	Vol Spikes	Extreme Drops	Volume Spikes	Anomaly Rate	IF Anomalies
RELIANCE	5,306	179	79	155	7.09%	~106
TCS	4,139	152	53	136	7.34%	~83
HDFCBANK	4,139	179	59	162	6.86%	~83
TATASTEEL	4,139	179	171	163	8.86%	~83

7.3 Crisis Validation

The 2008 GFC and 2020 COVID crash are clearly confirmed by anomaly clustering: RELIANCE shows 38 anomaly days in 2008 and 26 in 2020, both well above the annual average of ~18 days. TATASTEEL — a metals company with high commodity sensitivity — has the highest overall anomaly rate (8.86%) consistent with its exposure to global economic cycles.

The Isolation Forest multivariate approach captures anomalies that statistical thresholds alone might miss — particularly days where multiple indicators are simultaneously abnormal but none individually breaches the threshold.

8. Explainability & Transparency (Optional Task A)

Explainability is implemented at three levels: global model transparency, local per-prediction explanation, and quantitative justification of all portfolio and risk decisions.

8.1 SHAP — SHapley Additive exPlanations

SHAP (Lundberg & Lee, 2017) is integrated into both Notebook 06 and the Streamlit Explainable AI page. SHAP assigns each feature a contribution value for each specific prediction, grounded in cooperative game theory (Shapley values from economics).

SHAP Output	What It Shows	How It Is Used
Global Importance Bar Chart	Mean |SHAP| per feature across 300+ recent days	Identifies which indicators drive the model most consistently
Beeswarm Plot	Distribution of SHAP values showing direction and magnitude	Shows how high/low values of each feature shift predictions
Waterfall / Local Plot	Per-prediction breakdown for the most recent trading day	Explains exactly why today's signal is BUY or SELL
Dependence Plot	SHAP value vs feature value for top 4 features	Reveals non-linear relationships captured by the model
SHAP Over Time	Top feature's SHAP contribution plotted as time series	Shows how the model's reasoning evolves with market conditions

8.2 Portfolio Explainability

All portfolio allocations are derived from explicit quantitative objectives (minimise variance, maximise Sharpe, cap at target vol). The Efficient Frontier visualisation directly shows where each portfolio sits relative to 2,000 randomly sampled portfolios — communicating why each optimised portfolio is superior. Sector exposure pies and risk contribution charts show users the composition and concentration of their allocation.

8.3 Assumptions & Limitations

- Historical returns proxy expected returns — assumes partial stationarity. Structural regime breaks (e.g. post-COVID digitalisation) may reduce this assumption's validity.
- The sample covariance matrix is estimated over 2015–2021 and may underestimate tail correlations during crisis periods.
- Directional accuracy of 52–55% reflects genuine market unpredictability. Predictions are probabilistic guidance, not deterministic signals.
- Backtest results exclude transaction costs, slippage, and market impact — realised returns would be lower.
- All analysis uses data up to April 2021; market dynamics post-2021 are not captured.

9. Platform & Deployment (Optional Task E)

9.1 Streamlit Application — 8 Pages

Page	Key Features
🏠 Overview	Platform summary, sector breakdown bar chart, capability cards
🔍 Stock Analyser	Interactive candlestick + RSI + Volume + MACD, togglable indicators, 52-week stats, full risk profile
🤖 AI Predictor	Run ensemble model, probability gauge, BUY/SELL signal markers on price chart, all-stocks signal table with filters
💼 Portfolio Builder	4 profiles, weight charts, sector pie, performance vs benchmark, efficient frontier (2000 random portfolios), ₹ allocation slider
⚠️ Risk Assessment	VaR/CVaR histogram, rolling risk charts, portfolio comparison table, risk contribution bars, stress test grouped bar chart
📊 Market Overview	1-year return heatmap across all stocks, interactive correlation matrix, full risk scorecard table
🧠 Explainable AI	SHAP global importance, local waterfall explanation, SHAP over time — interactive, per-stock
🚨 Anomaly Detection	Anomaly overlay on price chart, vol/volume z-score panels, anomalies-by-year bar, top events table, universe scan

9.2 Performance & Caching

All data loading and model inference is wrapped in Streamlit's st.cache_data decorator. Model files are serialised with joblib and cached to the models/ directory (150 .pkl files for 49 stocks × 3 model components). First-time training for all 49 stocks takes approximately 8–12 minutes; all subsequent runs load instantly. The Efficient Frontier computation (2,000 random portfolios) runs in under 5 seconds due to vectorised NumPy operations.

10. Key Insights

10.1 Market Insights

- Average pairwise return correlation across the NIFTY-50 is 0.21 (2010–2021) — lower than expected, meaning cross-sector diversification provides meaningful but not complete risk reduction.
- BAJAJFINSV delivered a 2,972% total return over 10 years — the highest in the universe — reflecting compounding in the high-growth Indian financial services sector.
- Volatility exhibits strong clustering (ARCH effects), visible in rolling vol charts. This supports the use of ATR and HV_21 as ML features.
- COVID-19 produced the fastest recovery on record (~6 months vs ~24 months for GFC), reflecting increased institutional participation and RBI policy response.

10.2 Portfolio Insights

- Max Sharpe (Balanced) achieves Sharpe 0.95 vs 0.31 for Equal Weight — a 3× improvement through systematic Mean-Variance optimisation alone.
- During COVID crash, the Aggressive portfolio (−1.7%) outperformed Conservative (−12.3%) because IT and Consumer Goods — overweighted in the Aggressive profile — benefited from the digital acceleration narrative.
- Risk Parity offers the best diversification (Herfindahl index ~0.021 vs ~0.09 for Max Sharpe), at the cost of slightly lower absolute return.

10.3 Prediction Insights

- Return_1d (yesterday's return) is consistently the most predictive feature via SHAP — evidence of short-term momentum effects in NIFTY-50.
- Volatility regime (HV_21) is the second most important feature — stocks in low-vol regimes show more persistent directional trends.
- The ensemble (XGBoost + RF average) consistently outperforms either model alone by 0.5–1.5 percentage points in directional accuracy across the universe.

10.4 Anomaly Insights

- Anomaly rates cluster sharply in 2008 and 2020 across all stocks, validating both detection methods against known market crises.
- TATASTEEL has the highest anomaly rate (8.86%) — consistent with metals' sensitivity to global demand and commodity price cycles.
- Isolation Forest captures 2% of days as anomalous (contamination=0.02), identifying multivariate outliers that no single statistical threshold detects alone.

11. Conclusion

This project delivers a complete, end-to-end AI-powered investment intelligence platform for the NIFTY-50 universe. All three mandatory tasks are fully implemented: the Stock Predictor Engine (Task A) uses a walk-forward validated XGBoost + Random Forest ensemble; the Portfolio Construction Module (Task B) implements five distinct strategies including Markowitz optimisation and Risk Parity; and the Risk Assessment Module (Task C) provides comprehensive per-stock and portfolio-level risk analytics with stress testing across four historical crisis periods.

All four optional tasks are also addressed: Explainable AI (SHAP global and local explanations), Personalised Investment Strategies (three investor profiles), Market Anomaly Detection (statistical + Isolation Forest), and Deployment (8-page Streamlit web application).

The platform strictly adheres to competition constraints — only the provided NIFTY-50 Kaggle dataset is used, with no live market data, financial APIs, news data, or social media sentiment. Every feature is derived from OHLCV data alone.

The key contribution is integration: no single algorithm defines the platform. The combination of ensemble ML, Markowitz optimisation, historical risk analytics, SHAP explainability, and anomaly detection — all accessible through a unified interactive interface — creates a practical decision-support system that addresses the real challenge of transforming raw historical data into actionable investment intelligence.